

Who Discovers the Dollar?

Arbitrage Capacity and Price Discovery in Fragmented Stablecoin Markets

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Preliminary draft

Abstract

Identical dollar stablecoins trade at persistent, time-varying premia across local-currency markets, and the same order-flow imbalance can move one local premium a lot and another barely at all. We argue that the missing state variable is *arbitrage capacity*. When arbitrageurs face a proportional transfer cost and a convex balance-sheet cost, arbitrage is inactive inside a band and linear in the dislocation outside it. The premium then follows a threshold diffusion with a closed-form stationary law. The model implies that small premia do not revert while large ones do; that long-run price impact equals primitive impact divided by capacity, so volume reveals demand while the premium reveals capacity; that depletable capacity creates an arbitrage throughput limit beyond which the premium diverges; that the band filters which local information reaches the global price; and that hub-and-spoke arbitrage networks are efficient but fragile. Using 2024–2026 minute data on USDT against the Turkish lira, we find no mean reversion inside an estimated band of about 30 bp and significant reversion outside it. Arbitrage is four times faster during Istanbul banking hours, so the same order flow has nearly five times the long-run price impact at night. The stablecoin market does essentially all of the adjusting toward the interbank rate, and the market has become markedly more integrated over time. The dollar’s price is discovered in the interbank market; the stablecoin market pays for access to it.

Keywords: stablecoins, limits to arbitrage, market fragmentation, price discovery, order flow, threshold dynamics

JEL: G12, G14, G15, F31

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1 Introduction

Dollar stablecoins such as USDT are, by design, the same asset everywhere. Yet in local-currency markets (Turkish lira, Nigerian naira, Argentine peso, Korean won) the dollar-equivalent price of one USDT regularly deviates from parity by tens or hundreds of basis points, and these deviations can last for hours or days. The deviations are not simply a reflection of local demand. Two markets with similar signed order-flow imbalances can display very different premia, and a market with heavy trading volume can contribute little to the formation of the global dollar price. Volume, order flow and price impact, the standard microstructure statistics, appear to be only loosely connected.

This paper argues that the connection runs through a single latent object, *arbitrage capacity*, and it builds a model in which that object can be measured from high-frequency data. The premise is simple. Arbitrage between a local stablecoin market and the global dollar market is not frictionless and is not unlimited. Moving value across the two legs requires on- and off-ramps, banking rails, exchange withdrawals and, in some jurisdictions, access to foreign exchange. These frictions impose a per-unit cost. Arbitrageurs also have finite balance sheets and inventory tolerance, which makes large positions increasingly costly. We show that these two ingredients alone, a proportional cost c and a capacity parameter K that governs the convexity of balance-sheet costs, are enough to turn the textbook linear error-correction picture into a qualitatively different object with sharp, testable implications.

Mechanism. The arbitrageur chooses a flow a to maximize $ab - c|a| - a^2/(2K)$, where b is the log premium of the local market over the global benchmark. The solution is a soft-threshold rule,

$$a^*(b) = K \operatorname{sign}(b) (|b| - c)_+,$$

which is zero inside the band $[-c, c]$ and linear in the excess dislocation outside it. Feeding this policy into the premium dynamics, where non-arbitrage order flow q_t pushes the premium with impact β and arbitrage flow pushes it back with impact λ , gives a *threshold diffusion*:

$$db_t = \beta q_t dt - \lambda K \operatorname{sign}(b_t) (|b_t| - c)_+ dt + \sigma dW_t.$$

The noise term is not the point. The economic content sits in the drift, which is zero over an interval and has a kink at its edges. This differs from a model that simply replaces Brownian motion with a richer stochastic process.

Results. The model delivers a set of closed-form results.

(i) *Inaction and state-dependent reversion.* Inside the band the premium is a martingale: small dislocations do not revert and can persist indefinitely. Outside it, the excess decays at rate λK , with half-life $\ln 2/(\lambda K)$. The proportional speed of reversion, $\lambda K(1 - c/|b|)$, therefore *rises* with the size of the dislocation. Large mispricings correct faster than small ones. A linear error-correction model fitted to such data averages these speeds and gets both wrong.

(ii) *Volume reveals demand; the premium reveals capacity.* With persistent order flow q , the premium settles at $b^* = c + \beta q/(\lambda K)$. The long-run price impact of order flow is $\beta/(\lambda K)$, inversely proportional to capacity. Steady-state arbitrage volume, by contrast, equals $\beta q/\lambda$ and does not depend on K at all. Two markets with identical demand can therefore have identical arbitrage volume and very different premia, and two markets with identical premia can have very different demand. This turns the empirical puzzle of mismatched volume, order flow and premia into a statement about latent capacity.

(iii) *A closed-form stationary law.* With mean-zero order flow, the stationary density of the premium is flat on the band and Gaussian outside it, with tail precision $\lambda K/\sigma^2$. The band width, the capacities on each side and the volatility map one-to-one into the shape of the distribution. Asymmetric frictions ($c^+ \neq c^-$, $K^+ \neq K^-$) generate a non-zero average premium even when order flow has mean zero, so a persistent average premium is not, by itself, evidence of excess local demand.

(iv) *Capacity depletion and a throughput limit.* When arbitrage consumes capacity that replenishes at a finite rate, there is a critical order flow q_{crit} . Below it, long-run price impact is convex and explodes as $q \uparrow q_{crit}$. Above it, no steady state exists and the premium grows at least linearly in time. The maximal sustainable arbitrage flow, not the size of any one arbitrageur's balance sheet, is what bounds the market's ability to absorb demand. This captures, in a fragmented-market setting, the funding-liquidity spirals of Brunnermeier and Pedersen (2009) and the slow-moving capital of Duffie (2010).

(v) *Price discovery.* A local information shock of size δ reaches the global price only to the extent that it exceeds the band. By horizon h , the fraction incorporated is $\omega(1 - c/\delta)_+(1 - e^{-\lambda K h})$, where ω is the global leg's share of arbitrage price impact. Frictions determine *how much* local information is ever transmitted; capacity determines *how fast*. A segmented market can be very active and still discover very little of the dollar's value.

(vi) *Networks of venues.* With many venues linked by pairwise threshold arbitrage, the long-run expected cross-venue dispersion is bounded by a constant divided by the algebraic connectivity λ_2 of the capacity-weighted arbitrage graph. Among minimal (tree) arbitrage networks the hub-and-spoke star attains the largest λ_2 , so centralizing arbitrage through a hub maximizes normal-time integration. But removing the hub disconnects the network,

and dispersion then grows without bound. Centralized arbitrage is efficient and fragile at the same time.

Measurement. The model maps into a discrete-time threshold regression,

$$\Delta b_{t+1} = \alpha + \beta OFI_t - \kappa^+ (b_t - c^+)_{+} + \kappa^- (-b_t - c^-)_{+} + \varepsilon_{t+1}, \quad \kappa^{\pm} = \lambda K^{\pm},$$

which we estimate by profiling the sum of squared residuals over (c^+, c^-) . The structural objects of interest are the band (c^+, c^-) , the capacity-scaled reversion speeds $\kappa^{\pm}$, the primitive impact β and the long-run impact multiplier $\beta/\kappa^{\pm}$. The model also offers an identification device. Inside the band arbitrage flow is zero, so all observed order flow is non-arbitrage flow. In-band observations therefore identify the primitive impact β free of the mechanical contamination that arises when arbitrageurs' own trades are part of measured order flow. Monte Carlo evidence shows that the estimator recovers all parameters accurately at sample sizes of a few weeks of minute data. A linear error-correction model fitted to the same data understates the speed of arbitrage outside the band by a factor of two to eight across our designs, and it implies mean reversion inside the band, where none exists.

Evidence. We take the model to the market for USDT against the Turkish lira from January 2024 to August 2026. We combine one-minute Binance candles, which record taker-initiated volume and hence signed order flow, with one-minute interbank USD/TRY quotes. The premium of USDT over the interbank rate has a standard deviation of 56 bp. On the same exchange, the triangular basis against the BTC cross rate has a standard deviation of only 2.4 bp, so the relevant frictions sit in the flat leg of the arbitrage. Estimating month-specific bands, we find four results. First, the threshold structure is in the data. There is no detectable mean reversion inside a band of median width 32 bp, while the premium reverts significantly outside it, and linearity is rejected. Second, capacity varies with banking hours exactly as the model predicts. Above the band, arbitrage is four times faster during Istanbul business hours than outside them, while the primitive impact of order flow is statistically indistinguishable across the two states. A sustained order-flow imbalance therefore has a long-run price impact of about 56 bp during business hours and 257 bp outside them. Third, essentially all of the correction happens in the stablecoin price; the interbank rate does not respond to stablecoin order flow or to the premium. During the 19 March 2025 lira shock, the interbank rate jumped 11% while USDT/TRY rose 6%, opening a 660 bp discount that closed within 75 minutes. Fourth, between 2024 and 2026 the price impact of order flow fell by three quarters, the speed of arbitrage roughly doubled, and the volatility of the premium fell from 74 to 16 bp. Figure 1 summarizes the first two results. Together, these facts say

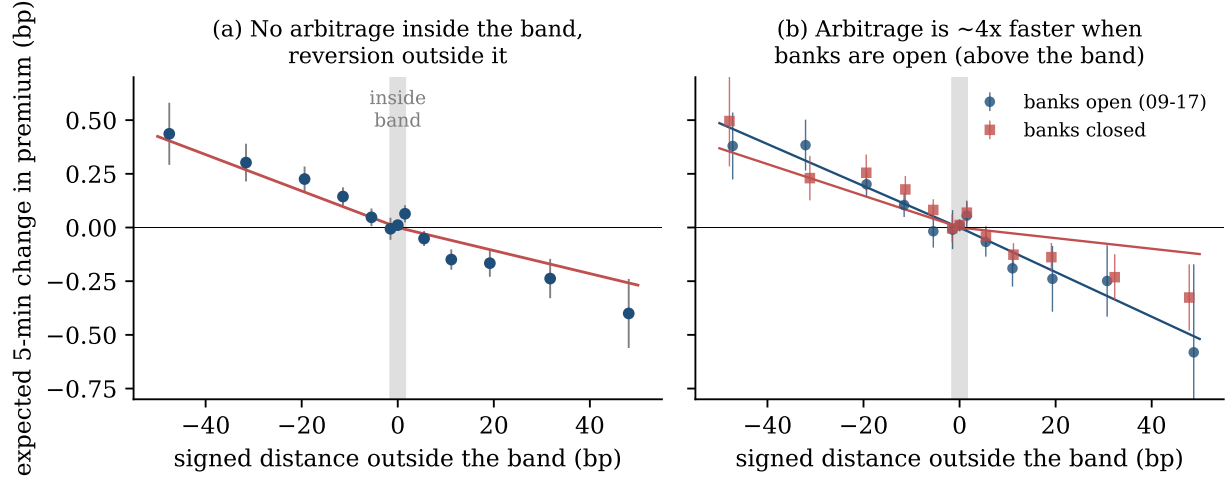


Figure 1: Arbitrage switches on outside a band, and switches on faster when banks are open. USDT/TRY premium over the interbank rate, five-minute intervals, January 2024 – August 2026. The horizontal axis is the signed distance of the premium outside its month-specific no-arbitrage band; all observations inside the band are plotted at zero (grey). The vertical axis is the expected change in the premium over the next five minutes, net of order flow and month fixed effects, with 95% confidence intervals clustered by day. Lines are the fitted threshold model (Table 2). Panel (a): full sample. Panel (b): Istanbul business hours versus other hours.

that the lira stablecoin market follows, rather than discovers, the dollar’s price, and that how closely it follows is governed by arbitrage capacity.

Related literature. The paper sits at the intersection of four literatures.

Cross-market crypto arbitrage. Makarov and Schoar (2020) document large and persistent cross-country deviations in cryptocurrency prices, link them to capital controls, and show that signed volume co-moves with spreads. Hautsch et al. (2024) show that settlement latency limits cross-exchange arbitrage, and Griffin and Shams (2020) study the interaction of stablecoin issuance and crypto prices. Hu and Zhang (2023) model how arbitrageurs endogenously connect exchanges and show that star-shaped arbitrage networks can arise in equilibrium. We take the network as given and ask what its topology implies for convergence and fragility. Our contribution relative to this literature is a microstructure model in which a single state variable, arbitrage capacity, governs both the mapping from order flow to the premium and the transmission of local information to the global price, with closed-form, testable implications.

Stablecoin stability and arbitrage. Lyons and Viswanath-Natraj (2023) study the arbitrage that keeps stablecoins close to their peg. Ma et al. (2025) show that concentrating primary-market arbitrage in a few authorized intermediaries trades off price stability in normal times against run risk. Our network result is the secondary-market, cross-venue analogue of their

trade-off. [Aldasoro et al. \(2026\)](#) show that stablecoin inflows spill over into FX markets and widen parity deviations, and they interpret the non-linearity through constrained arbitrage. We provide a high-frequency, microstructure foundation for that interpretation and a way to measure the constraint directly.

Limits to arbitrage and slow-moving capital. The arbitrageur problem builds on [Shleifer and Vishny \(1997\)](#), [Gromb and Vayanos \(2002\)](#), [Brunnermeier and Pedersen \(2009\)](#), [Mitchell et al. \(2007\)](#) and [Duffie \(2010\)](#). Deviations from covered interest parity ([Du et al., 2018](#)) are a leading example in traditional markets of arbitrage-capacity-driven basis. The band structure echoes the commodity-points literature in international economics ([Dumas, 1992](#); [Obstfeld and Taylor, 1997](#)) and threshold cointegration ([Balke and Fomby, 1997](#); [Hansen and Seo, 2002](#)). We derive the threshold from an arbitrageur’s optimization, connect it to order flow, and characterize its implications for price discovery and network fragility.

Fragmentation and price discovery. Our price-discovery result complements the information-share approach of [Hasbrouck \(1995\)](#) and [Gonzalo and Granger \(1995\)](#), which is built on linear error correction. Under threshold arbitrage, the contribution of a venue to price discovery depends on the size of the information it receives, so linear information shares average over economically distinct regimes. [Lehar et al. \(2024\)](#) show how fixed costs fragment liquidity across fee tiers within a decentralized exchange. We study fragmentation across venues and currencies and focus on its implications for convergence.

The remainder of the paper is organized as follows. Section 2 presents the model. Section 3 derives its implications. Section 4 studies networks of venues. Section 5 develops the estimator and the empirical design. Section 6 presents the evidence from the lira market. Section 7 reports simulation evidence on the estimator. Section 8 concludes. Proofs are in the Appendix.

2 Model

2.1 Markets and the premium

There are two markets for the same dollar stablecoin. In the *local* market the stablecoin trades against a local currency; in the *global* market it trades against the U.S. dollar (or a dollar benchmark such as a deep USDT/USD or USDC/USDT book). Let P_t^L denote the local price converted into dollars at a reference exchange rate, and P_t^G the global dollar price. The object of interest is the log premium

$$b_t \equiv \log P_t^L - \log P_t^G.$$

In a frictionless integrated market $b_t \equiv 0$. Which exchange rate is used for the conversion determines the economic interpretation. Converting at the official rate makes b_t a measure of the parallel-market premium, while converting at a contemporaneous local market rate (for example, a local BTC/USDT cross) isolates the stablecoin-specific basis. The model applies to either.

Time is continuous. Three forces move the premium.

Non-arbitrage order flow. End users (households seeking dollar exposure, importers, remitters) generate signed order flow q_t in the local market, measured as buyer-initiated minus seller-initiated volume. Absent arbitrage, order flow moves the premium with primitive impact $\beta > 0$.

Arbitrage flow. A competitive arbitrage sector chooses a flow a_t (units per unit time). A positive a_t sells the stablecoin locally and buys it globally, which lowers the premium. Arbitrage moves the local price with impact λ_L and the global price with impact λ_G , so its effect on the premium is $\lambda \equiv \lambda_L + \lambda_G$.

Noise. Everything else, including transient liquidity shocks and quote noise, is captured by σdW_t , where W is a standard Brownian motion.

2.2 The arbitrage sector

Assumption 1 (Arbitrage technology). Arbitrage in the positive (negative) direction incurs a proportional cost $c^+ > 0$ ($c^- > 0$) per unit, reflecting fees, on/off-ramp costs, FX access and transfer frictions. Holding a flow of size $|a|$ incurs a convex balance-sheet and inventory cost $a^2/(2K^\pm)$, where $K^\pm > 0$ is the *arbitrage capacity* in each direction. The arbitrage sector chooses a_t to maximize flow profit

$$\Pi(a; b) = \begin{cases} ab - c^+a - \frac{a^2}{2K^+}, & a \geq 0, \\ ab + c^-a - \frac{a^2}{2K^-}, & a < 0. \end{cases} \quad (1)$$

The proportional cost $c^\pm$ captures every friction that scales with the size of the trade: exchange and network fees, the spread paid on the fiat leg, and the expected cost of on- and off-ramps. Capacity $K^\pm$ captures everything that makes the *marginal* unit costlier: finite capital, inventory risk while transfers settle, counterparty limits at banks and payment providers, and withdrawal limits at exchanges. Allowing both to depend on direction is essential in capital-controlled economies. Buying stablecoins with local currency and moving the dollars abroad is generally harder than the reverse, so one expects $c^+ \neq c^-$ and $K^+ \neq K^-$.

Proposition 1 (Optimal arbitrage). *Under Assumption 1, the unique optimal arbitrage flow is*

$$a^*(b) = K^+ (b - c^+)_+ - K^- (-b - c^-)_+. \quad (2)$$

In particular, $a^(b) = 0$ for $b \in [-c^-, c^+]$; outside this no-arbitrage band, arbitrage flow is linear in the excess premium with slope equal to capacity. In the symmetric case, $a^*(b) = K \operatorname{sign}(b) (|b| - c)_+$.*

The policy is the soft-thresholding operator familiar from statistics. Its two parameters play distinct economic roles. The friction c determines *whether* arbitrage is active; capacity K determines *how strongly* it responds once active. Figure 2(a) plots the policy.

Remark 1 (Static versus dynamic arbitrage). Proposition 1 treats arbitrage as a flow decision. In a fully dynamic model with fixed costs, the option value of waiting widens the band beyond the static cost, as in Dumas (1992). The structure of the solution (inaction inside a band, active trading outside it) is preserved. The band $c^\pm$ estimated in Section 5 should therefore be read as an *effective* threshold that includes any such option value. More generally, for any convex cost $C(a)$ with $C'(0^+) = c^+$ and $C'(0^-) = -c^-$, the optimal policy is $a^*(b) = (C')^{-1}(b)$ outside $[-c^-, c^+]$ and zero inside, so all qualitative results below survive. Only the closed forms rely on the quadratic specification.

Remark 2 (Settlement latency and the location of the band). Cross-border stablecoin arbitrage is not instantaneous. Fiat legs settle with a delay, exchanges can hold withdrawals, and transfers may be subject to compliance reviews. Suppose a round trip ties up capital for a period τ , during which the arbitrageur holds dollar-denominated inventory rather than local-currency deposits. The arbitrageur then forgoes the interest differential $(i - i^*)$ over τ in *both* directions, so the band becomes $[-c^- - (i - i^*)\tau, c^+ + (i - i^*)\tau]$. In a high-interest-rate economy this term is large. At a 45% annual lira rate, a 48-hour withdrawal hold adds roughly 25 bp to each edge of the band. Directional frictions, such as limits on outbound stablecoin transfers, raise only one edge. They therefore shift the band's location and, by Proposition 4(b), the average premium. Empirically, we estimate the edges $\ell \equiv -c^-$ and $u \equiv c^+$ freely, so the band need not contain zero.

2.3 Premium dynamics

Combining non-arbitrage order flow, optimal arbitrage and noise, the premium satisfies

$$db_t = \left(\beta q_t - \lambda \left[K^+ (b_t - c^+)_+ - K^- (-b_t - c^-)_+ \right] \right) dt + \sigma dW_t. \quad (3)$$

We write $\kappa^\pm \equiv \lambda K^\pm$ for the capacity-scaled reversion speeds and

$$\mu(b) \equiv -\kappa^+ (b - c^+)_+ + \kappa^- (-b - c^-)_+$$

for the arbitrage component of the drift. Equation (3) is a threshold diffusion. Its drift is identically zero on the band, linear outside it, and kinked at $-c^-$ and c^+ . Since μ is Lipschitz, (3) has a unique strong solution for any adapted, locally integrable order-flow process q .

3 Implications

3.1 Inaction and state-dependent reversion

Proposition 2 (State-dependent reversion). *Let $q \equiv 0$.*

- (a) *Inaction. For $b_0 \in (-c^-, c^+)$, let τ be the first exit time from the band. The stopped premium $b_{t \wedge \tau}$ is a martingale, and*

$$\mathbb{E}[\tau \mid b_0] = \frac{(c^+ - b_0)(b_0 + c^-)}{\sigma^2}.$$

- (b) *Exponential correction outside the band. If $\sigma = 0$ and $b_0 > c^+$, then $b_t = c^+ + (b_0 - c^+)e^{-\kappa^+ t}$. The excess premium has half-life*

$$t_{1/2}^+ = \frac{\ln 2}{\lambda K^+},$$

and symmetrically $t_{1/2}^- = \ln 2 / (\lambda K^-)$ for $b_0 < -c^-$.

- (c) *Large dislocations revert faster. The proportional reversion speed $\rho(b) \equiv -\mu(b)/b$ equals zero on the band and $\kappa^+(1 - c^+/b)$ for $b > c^+$ (symmetrically for $b < -c^-$). It is non-decreasing in $|b|$ and converges to $\kappa^\pm$ as $|b| \rightarrow \infty$.*

Proposition 2 contains the first set of testable predictions. Small premia do not mean-revert; they wander until they hit the edge of the band, with an expected waiting time that rises with the square of the band width. Large premia revert, and do so proportionally faster the larger they are. A linear error-correction model imposes a constant ρ . Figure 2(b) shows that any constant necessarily overstates reversion for small dislocations and understates it for large ones. The half-life formula gives capacity a direct, observable counterpart: halving capacity doubles the half-life of a dislocation.

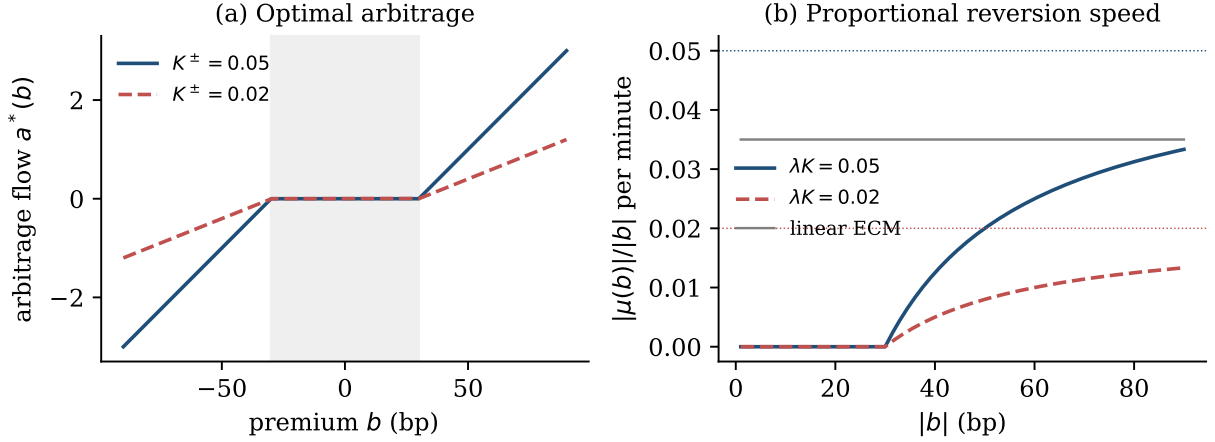


Figure 2: Optimal arbitrage and state-dependent reversion. Panel (a) plots the optimal arbitrage flow $a^*(b)$ of Proposition 1 for two levels of capacity with $c^\pm = 30$ bp; the shaded region is the no-arbitrage band. Panel (b) plots the proportional reversion speed $|\mu(b)|/|b|$ of Proposition 2(c), with $\lambda = 1$; dotted lines mark the asymptotes $\kappa = \lambda K$. The flat grey line is a constant-speed linear error-correction model.

3.2 Order flow, price impact and volume

Proposition 3 (Long-run price impact). *Let $\sigma = 0$ and $q_t \equiv q$ constant.*

(a) *If $q > 0$, then from any initial condition $b_t \rightarrow b^*$, where*

$$b^* = c^+ + \frac{\beta q}{\lambda K^+}.$$

If $q < 0$, $b_t \rightarrow -c^- + \beta q/(\lambda K^-)$. If $q = 0$, b_t converges to the closest point of the band.

(b) *The long-run price impact of order flow is*

$$\frac{\partial b^*}{\partial q} = \frac{\beta}{\lambda K^\pm},$$

which is decreasing in capacity.

(c) *In the steady state, arbitrage flow equals $a^* = \beta q/\lambda$, which does not depend on $(c^\pm, K^\pm)$. Total local volume, $|q| + |a^*|$, is therefore independent of arbitrage capacity, while the premium is decreasing in it.*

Part (c) is the paper's central observational-equivalence result. In steady state arbitrageurs must absorb the full non-arbitrage flow, scaled by relative impact, whatever their capacity. Capacity determines only the premium at which they are willing to do so. Volume reveals demand; the premium reveals capacity.

Corollary 1 (Volume, premia and demand across markets). *Consider two local markets $i \in \{A, B\}$ with parameters $(\beta_i, \lambda_i, c_i, K_i)$ and persistent order flow $q_i > 0$. Then*

$$b_A^* - c_A = b_B^* - c_B \iff \frac{\beta_A q_A}{\lambda_A K_A} = \frac{\beta_B q_B}{\lambda_B K_B}.$$

In particular, equal excess premia are consistent with arbitrarily different demand, provided capacity differs in proportion. Conversely, equal demand and impact ($\beta_A q_A = \beta_B q_B$, $\lambda_A = \lambda_B$) produce excess premia in inverse proportion to capacity.

Corollary 1 explains why comparisons of premia across countries, or across episodes within a country, cannot by themselves be read as comparisons of demand. Separating demand from capacity requires the joint dynamics of the premium and order flow, which is exactly what the estimator of Section 5 uses.

3.3 The stationary distribution of the premium

When order flow has mean zero and is short-lived relative to arbitrage, its effect is absorbed into the diffusion term. The premium then has an explicit stationary law.

Proposition 4 (Stationary distribution). *Let $q \equiv 0$. The premium process (3) is positive recurrent with unique stationary density*

$$\pi(b) = \frac{1}{Z} \begin{cases} \exp\left[-\frac{\lambda K^+}{\sigma^2}(b - c^+)^2\right], & b > c^+, \\ 1, & -c^- \leq b \leq c^+, \\ \exp\left[-\frac{\lambda K^-}{\sigma^2}(b + c^-)^2\right], & b < -c^-, \end{cases} \quad (4)$$

with normalizing constant

$$Z = c^+ + c^- + s^+ + s^-, \quad s^\pm \equiv \frac{1}{2} \sqrt{\frac{\pi \sigma^2}{\lambda K^\pm}}.$$

Consequently:

- (a) *The stationary probability that the premium lies inside the band is $(c^+ + c^-)/Z$.*
- (b) *The stationary mean premium is*

$$\mathbb{E}_\pi[b] = \frac{1}{Z} \left[\frac{(c^+)^2 - (c^-)^2}{2} + c^+ s^+ - c^- s^- + \frac{\sigma^2}{2\lambda K^+} - \frac{\sigma^2}{2\lambda K^-} \right],$$

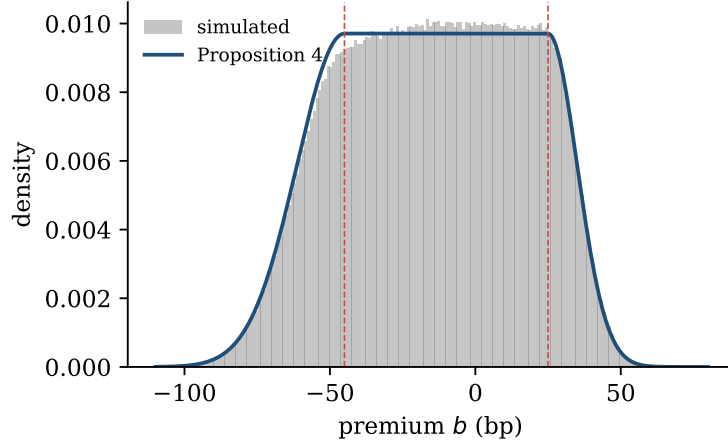


Figure 3: Stationary distribution of the premium. Asymmetric design with $c^+ = 25$, $c^- = 45$ bp, $\lambda K^+ = 0.08$, $\lambda K^- = 0.03$ per minute and $\sigma = 4$ bp/ $\sqrt{\text{min}}$. The histogram is a simulated path of 2×10^6 minutes; the solid line is the closed form (4). Dashed lines mark the band. The theoretical probability of being inside the band is 0.680, against 0.688 in the simulation (the small gap reflects Euler discretization at the boundary).

which is non-zero whenever frictions or capacities are asymmetric, even though order flow has mean zero.

The density in (4) has a distinctive shape: a flat plateau whose width is the band, joined continuously to half-Gaussian tails whose precision $\lambda K^\pm / \sigma^2$ measures capacity relative to noise. Figure 3 compares it with a long simulated path. Part (b) cautions against reading a persistent average premium as excess demand. If it is harder to move dollars out of the country than in ($c^+ > c^-$ or $K^+ < K^-$), the premium is positive on average even when order flow is balanced.

3.4 Capacity depletion and the arbitrage throughput limit

Capacity is not a fixed endowment. Arbitrage ties up capital in transit, exhausts withdrawal limits and banking lines, and accumulates inventory that must be unwound. We now let capacity respond to arbitrage activity. Focus on the region $b > c^+$ and drop direction superscripts:

$$dK_t = [\xi(\bar{K} - K_t) - \chi |a_t^*|]dt, \quad a_t^* = K_t(b_t - c)_+, \quad (5)$$

where $\bar{K}$ is normal-times capacity, ξ is the rate at which capacity replenishes (capital is redeployed, limits reset), and χ is the rate at which arbitrage activity consumes it.

Proposition 5 (Throughput limit and convex impact). *Let $\sigma = 0$, $q_t \equiv q > 0$, and let (b_t, K_t) follow (3) and (5) with $K_0 > 0$. Define the arbitrage throughput limit $\bar{a} \equiv \xi \bar{K} / \chi$*

and the critical order flow

$$q_{crit} \equiv \frac{\lambda \bar{a}}{\beta} = \frac{\xi \lambda \bar{K}}{\chi \beta}.$$

(a) If $q < q_{crit}$, there is a unique steady state,

$$K^* = \bar{K} \left(1 - \frac{q}{q_{crit}}\right), \quad b^* = c + \frac{\beta q}{\lambda \bar{K} (1 - q/q_{crit})},$$

and it is locally asymptotically stable. The long-run price impact

$$\frac{\partial b^*}{\partial q} = \frac{\beta}{\lambda \bar{K} (1 - q/q_{crit})^2}$$

is increasing and convex in q and diverges as $q \uparrow q_{crit}$.

(b) If $q > q_{crit}$, no steady state exists, and for any initial condition

$$\liminf_{t \rightarrow \infty} \frac{b_t}{t} \geq \beta (q - q_{crit}) > 0.$$

Proposition 5 identifies the binding constraint in stressed markets. The constraint is not the stock of capacity K but the flow $\bar{a} = \xi \bar{K} / \chi$, the rate at which the arbitrage sector can sustainably process trades. Demand above this throughput limit cannot be absorbed at any premium, and the premium drifts away. This is the mechanism behind sudden depegs of local stablecoin prices during currency crises and capital-control tightening. Demand surges, arbitrage turns on, capacity is consumed, and future corrections slow down, which lets the premium keep widening. Figure 4(a) shows the convex steady-state impact; panel (b) shows the response to a temporary burst of order flow. In the illustrative calibration, depletion doubles the peak premium and raises the half-life of the subsequent correction from 14 minutes to more than six hours.

3.5 Price discovery: who discovers the dollar?

We now ask how information that arrives first in the local market is transmitted to the global price. Let the local price jump by $\delta > 0$ at $t = 0$ because informed local order flow reveals news about the common dollar value (for example, news about the issuer, or about the dollar's local purchasing power that global participants learn only through prices). Before the shock $b_{0-} = 0$. Arbitrage moves the local price by $-\lambda_L a dt$ and the global price by $+\lambda_G a dt$. Define the global leg's share of arbitrage impact $\omega \equiv \lambda_G / (\lambda_L + \lambda_G)$.

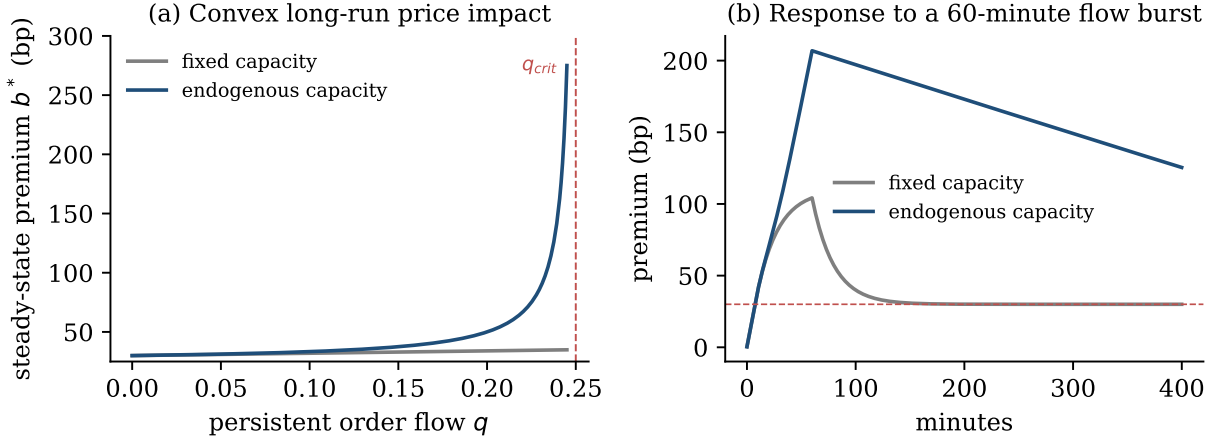


Figure 4: Capacity depletion. Parameters: $\beta = \lambda = 1$, $\bar{K} = 0.05$, $\xi = 0.01$, $\chi = 0.002$, $c = 30$ bp, so $q_{crit} = 0.25$. Panel (a): steady-state premium against persistent order flow, with fixed capacity (grey) and endogenous capacity (Proposition 5). Panel (b): response to order flow $q = 4$ for 60 minutes. With fixed capacity the post-burst half-life of the excess premium is 13.9 minutes, equal to $\ln 2/(\lambda \bar{K})$; with endogenous capacity it is 370 minutes.

Proposition 6 (Local information transmission). *Let $\sigma = 0$ and $q_t \equiv 0$ for $t > 0$. The global price at horizon h satisfies*

$$S_h(\delta) \equiv \frac{\log P_h^G - \log P_0^G}{\delta} = \omega \left(1 - \frac{c^+}{\delta} \right)_+ \left(1 - e^{-\lambda K^+ h} \right).$$

Consequently:

- (a) *local information shocks smaller than the band ($\delta \leq c^+$) are never transmitted to the global price;*
- (b) *the long-run share $S_\infty(\delta) = \omega(1 - c^+/\delta)_+$ is decreasing in the friction and independent of capacity;*
- (c) *at any finite horizon, $S_h(\delta)$ is increasing in capacity, and the half-life of transmission is $\ln 2/(\lambda K^+)$.*

Frictions determine how much local information ever reaches the global price; capacity determines how fast it gets there (Figure 5). A market with heavy trading but a wide band, or one whose arbitrage capacity is impaired, can be very active and still contribute almost nothing to the price of the dollar elsewhere. This is consistent with the finding of [Makarov and Schoar \(2020\)](#) that more segmented, higher-premium venues contribute less to global price discovery, and it explains that finding through (c, K) . A further implication concerns measurement. Linear information-share measures ([Hasbrouck, 1995](#); [Gonzalo and Granger,](#)

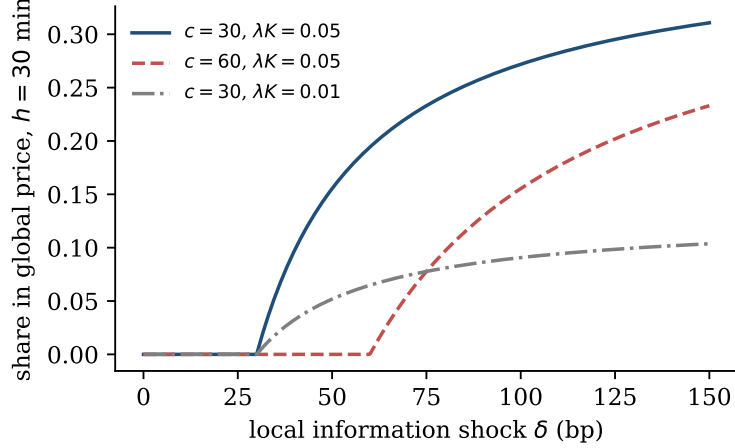


Figure 5: Transmission of local information to the global price. Fraction of a local information shock of size δ incorporated into the global price within $h = 30$ minutes (Proposition 6), with $\omega = 0.5$, for different frictions c and capacities λK .

1995) average $S_h(\delta)$ over the distribution of shock sizes. They therefore depend on the volatility regime and can change sharply with no change in market structure.

4 Networks of venues

Stablecoins trade on many venues at once: several local exchanges, peer-to-peer platforms, and global order books. We now let $N \geq 3$ venues with log prices $p_t = (p_{1t}, \dots, p_{Nt})'$ be connected by pairwise threshold arbitrage. Let $w_{ij} = w_{ji} \geq 0$ be the capacity-scaled arbitrage intensity on link $\{i, j\}$, and $c_{ij} = c_{ji} > 0$ its friction. Let $\phi_c(x) \equiv \text{sign}(x) (|x| - c)_+$ denote the soft-threshold function. Prices follow

$$dp_{it} = - \sum_{j \neq i} w_{ij} \phi_{c_{ij}}(p_{it} - p_{jt}) dt + (\Sigma dW_t)_i, \quad (6)$$

where Σ is an $N \times N$ volatility matrix that also absorbs mean-zero order-flow shocks. Let L_w be the weighted graph Laplacian of (w_{ij}) , and let $\lambda_2(L_w)$ be its second-smallest eigenvalue, the *algebraic connectivity* of the arbitrage network (Fiedler, 1973). The network is connected if and only if $\lambda_2(L_w) > 0$. Cross-venue disagreement is measured by $z_t = Pp_t$ with $P = I - \frac{1}{N}\mathbf{1}\mathbf{1}'$ and $V(z) = \frac{1}{2}\|z\|^2$.

Proposition 7 (Dispersion and connectivity). *If the arbitrage network is connected, then*

$$\limsup_{t \rightarrow \infty} \mathbb{E}[V(z_t)] \leq \frac{\frac{1}{2} \sum_{i < j} w_{ij} c_{ij}^2 + \frac{1}{2} \text{tr}(P \Sigma \Sigma' P)}{\lambda_2(L_w)}.$$

If the network is disconnected, $\mathbb{E}[V(z_t)]$ grows linearly in t whenever the noise is non-degenerate across components.

The bound has two terms in the numerator. The first is the dispersion that frictions tolerate inside the bands; the second is the dispersion generated by noise. The denominator is the network’s ability to pull prices together. Long-run dispersion therefore depends on capacity only through λ_2 , which depends on the *topology* of the network as well as on the level of capacity. Two corollaries follow.

Corollary 2 (Centralized arbitrage: efficient and fragile). *Suppose links are costly, so that the arbitrage network is a spanning tree, and each link has the same intensity w .*

- (a) *The star network, in which a single hub venue is linked to every other venue, attains $\lambda_2(L_w) = w$. This is the maximum over all spanning trees, so the hub structure minimizes the dispersion bound among minimal networks.*
- (b) *If the hub fails, all links vanish, the remaining $N - 1$ venues are mutually disconnected, and dispersion grows without bound. Removing any single non-hub venue leaves $\lambda_2 = w$ unchanged.*
- (c) *By contrast, in the complete network with intensity w on every link, $\lambda_2 = Nw$, and removing any venue leaves $\lambda_2 = (N - 1)w$.*

Corollary 2 formalizes a trade-off. When maintaining an arbitrage link is costly (accounts, banking relationships and transfer rails must be set up for each pair of venues), arbitrage naturally concentrates in a hub. In crypto the hub is typically the largest global exchange or a dominant market maker (Hu and Zhang, 2023). The hub maximizes integration in normal times, but its failure disconnects the market. This mirrors, across secondary-market venues, the primary-market trade-off documented by Ma et al. (2025), and it predicts that the dispersion of local stablecoin prices should spike when a hub venue suspends withdrawals or a hub arbitrageur exits, by much more than when a peripheral venue does.

5 Empirical implementation

5.1 Measurement

Premium. For each local market (e.g., USDT/TRY, USDT/NGN), we construct the premium at a one-minute frequency,

$$b_t = \log M_t^L - \log X_t,$$

where M_t^L is the local mid-quote of USDT in local currency and X_t is a contemporaneous benchmark price of one U.S. dollar in local currency. Two benchmarks are informative: an interbank or official FX rate, and a crypto-implied rate (the local BTC price divided by the global BTC/USDT price). The difference between the two premia separates FX-access frictions from crypto-venue frictions.

Order flow. Signed order-flow imbalance is constructed from trade-level data as buyer-initiated minus seller-initiated volume in USDT. Where only order-book snapshots are available, we use the order-flow imbalance of [Cont et al. \(2014\)](#). We standardize OFI_t within market to make β comparable across markets.

Capacity states. The model predicts that $c^\pm$ and $K^\pm$ vary with observable states. We consider (i) banking hours versus nights and weekends, when fiat rails are closed and K falls mechanically; (ii) exchange withdrawal suspensions and on-ramp outages; (iii) regulatory announcements and enforcement actions targeting crypto on- and off-ramps; (iv) volatility of the global stablecoin market; and (v) the cumulative recent arbitrage flow, which proxies the depletion term in (5).

5.2 Estimator

An Euler discretization of (3) at interval Δ gives

$$\Delta b_{t+1} = \alpha + \beta OFI_t - \kappa^+ (b_t - c^+)_{+} + \kappa^- (-b_t - c^-)_{+} + \varepsilon_{t+1}, \quad (7)$$

with $\kappa^\pm = \lambda K^\pm \Delta$. For fixed (c^+, c^-) , (7) is linear in $(\alpha, \beta, \kappa^+, \kappa^-)$. We estimate by profiled least squares:

$$(\hat{c}^+, \hat{c}^-) = \arg \min_{(c^+, c^-) \in \mathcal{G}} \text{SSR}(c^+, c^-),$$

over a grid $\mathcal{G}$, and then obtain the slopes by OLS at $(\hat{c}^+, \hat{c}^-)$. Thresholds in such models are super-consistent ([Hansen, 2000](#)), so inference on the slopes can treat them as known. Confidence sets for the thresholds are obtained by inverting the likelihood-ratio statistic as in [Hansen \(2000\)](#). The null of linear error correction (no band) is tested with the sup-LR statistic and the bootstrap of [Hansen \(1996\)](#) and [Hansen and Seo \(2002\)](#). Capacity states Z_t enter through interactions, $\kappa_t^\pm = \kappa_0^\pm + \kappa_1^{\pm'} Z_t$ and $c_t^\pm = c_0^\pm + c_1^{\pm'} Z_t$; the latter are estimated by splitting the sample on discrete states.

Structural objects. The estimator delivers, market by market and state by state:

- the band (c^+, c^-) , and its asymmetry, which measures the difference between inbound and outbound frictions;
- capacity-scaled speeds $\kappa^\pm$ and half-lives $\ln 2/\kappa^\pm$;
- the primitive impact β ;
- the long-run impact multiplier $\beta/\kappa^\pm$ (Proposition 3), which converts a unit of persistent order flow into a steady-state premium.

Capacity K and arbitrage impact λ are identified only through their product. Cross-market comparisons of κ are comparisons of capacity under the maintained assumption that λ is comparable. This can be checked using the depth of the two legs.

5.3 Identification of order-flow impact

Measured order flow contains the arbitrageurs' own trades, which are mechanically correlated with the premium: when $b_t > c^+$, arbitrageurs sell locally, so measured OFI_t is pushed down exactly when the premium is about to fall. OLS of (7) on total order flow therefore biases β towards zero. The model suggests a natural remedy. By Proposition 1, arbitrage flow is zero inside the band, so in-band order flow is pure non-arbitrage flow. Estimating β on the subsample $b_t \in [\hat{c}^-, \hat{c}^+]$ (or instrumenting total order flow with in-band order flow) recovers the primitive impact. Complementary strategies use trade-size or account-type filters to isolate retail flow, and the timing restriction that lagged order flow is predetermined with respect to ε_{t+1} .

5.4 Tests

The model yields the following tests, in order of increasing structure.

- T1.** *Inaction region.* The reversion coefficient is zero for $|b_t|$ below the estimated band and positive above it; the sup-LR test rejects linear error correction.
- T2.** *Size-dependent speed.* In regressions of $-\Delta b_{t+1} \text{sign}(b_t)$ on $|b_t|$ within bins of $|b_t|$, the slope is non-decreasing across bins (Proposition 2(c)).
- T3.** *Asymmetry.* $c^+ \neq c^-$ and $\kappa^+ \neq \kappa^-$, with the direction predicted by the asymmetry of capital controls.
- T4.** *Capacity states.* Bands widen and κ falls outside banking hours, during withdrawal suspensions and after regulatory tightening; the long-run impact multiplier β/κ rises correspondingly.

- T5. *Depletion.*** Conditional on b_t , κ is lower after periods of heavy cumulative arbitrage flow, and long-run impact is convex in persistent order flow (Proposition 5).
- T6. *Price discovery.*** The response of the global price to a local premium innovation is zero for innovations inside the band and increasing in the excess outside it, and the transmission speed is increasing in estimated κ (Proposition 6).

6 Evidence from the lira stablecoin market

We now take the model to the market for USDT against the Turkish lira, one of the largest local-currency stablecoin markets in the world. Turkey is a natural laboratory. Domestic demand for dollar assets is persistent and strong. The lira was hit by a large political shock during the sample. Policy interest rates were very high, so the carry cost of settlement latency in Remark 2 is economically large. Access to the fiat leg of the arbitrage runs through domestic banking rails whose availability varies by time of day.

6.1 Data and construction

Local market. We use one-minute USDT/TRY candles from Binance’s public data archive, from 2 January 2024 to 31 August 2026. For each minute the archive reports volume, the number of trades, quote volume and *taker-buy* volume. The local price P_t^L is the minute volume-weighted average price (quote volume divided by volume). Signed order flow is taker-buy minus taker-sell volume, $OFI_t = 2TBV_t - V_t$, in USDT. The market is liquid: the median minute has 90 trades and 36 thousand USDT of volume (Table 1).

Benchmark. The global dollar price is the interbank USD/TRY mid-quote from Dukascopy one-minute bid and ask candles. Its median relative bid-ask spread is about 1 bp. We use minutes with interbank activity only, exclude weekends and the 20:30–22:30 UTC rollover window (when interbank quotes are stale or very wide), and discard the handful of minutes with $|b_t| > 10\%$. The premium is $b_t = 10^4(\log P_t^L - \log P_t^G)$ in basis points.

Sampling. We average the premium over non-overlapping five-minute intervals, which removes most of the bid-ask bounce in the local price, and sum order flow over the same interval. We standardize order flow by its within-month standard deviation, so that β is the premium response in basis points to a one-standard-deviation five-minute order-flow shock.

Table 1: Summary statistics: USDT/TRY, January 2024 – August 2026

	Mean	SD	P1	P50	P99	N
Premium b_t (bp), 1-min	18.43	56.43	-60.06	5.76	269.96	808,473
Premium (bp), 5-min average	18.30	56.34	-60.22	5.62	269.74	162,624
USDT/TRY volume (thousand USDT per min)	64.91	129.97	1.13	36.42	543.61	808,473
Signed order flow (thousand USDT per min)	9.12	112.69	-212.43	2.46	210.12	808,473
Trades per minute	143.40	204.96	8.00	90.00	952.00	808,473
Interbank USD/TRY bid-ask spread (bp)	1.26	1.09	0.50	0.97	6.74	15,149
Triangular basis USDT/TRY vs. BTC cross (bp)	-0.13	2.38	-6.18	-0.10	5.68	1,378,099

Notes: One-minute data on weekdays, excluding 20:30–22:30 UTC. The premium is the log difference between the Binance USDT/TRY minute VWAP and the Dukascopy interbank USD/TRY mid. Signed order flow is taker-buy minus taker-sell volume. The triangular basis compares USDT/TRY with the cross rate implied by BTC/TRY and BTC/USDT on the same exchange (all days).

The regression is

$$\Delta b_t = \alpha_m + \beta OFI_t - \kappa^+ (b_{t-1} - u_m)_+ + \kappa^- (\ell_m - b_{t-1})_+ + \varepsilon_t, \quad (8)$$

where t indexes five-minute intervals, order flow is measured over the same interval as the price change, α_m is a month fixed effect, and $[\ell_m, u_m]$ is a month-specific band. The final sample has 162,624 intervals over 694 trading days.

Why month-specific bands. The premium’s level moves substantially at low frequency (Figure 6). It averaged more than 100 bp in the first half of 2024, fluctuated around zero in 2025, and turned into a discount of 10–25 bp in 2026. Remark 2 and Proposition 4(b) attribute such movements to slow-moving carry and directional frictions, which shift the band. We therefore estimate the band edges month by month. In each month, (ℓ_m, u_m) are profiled over a 1 bp grid subject to $\ell_m \leq u_m$ and 5% trimming. Given the bands, we estimate the common slopes in (8) by pooled OLS. Thresholds are super-consistent (Hansen, 2000), so slope inference treats them as known. All standard errors are clustered by day.

The friction is in the fiat leg. Table 1 contains a useful benchmark. On the same exchange, the USDT/TRY price and the USDT price implied by the BTC/TRY and BTC/USDT order books differ by only 2.4 bp (standard deviation). This triangular arbitrage requires no movement of fiat money and is essentially frictionless. The premium over the interbank rate, by contrast, has a standard deviation of 56 bp. The frictions that matter for the dollar premium are those of the fiat and cross-border leg, which is what c and K measure in the model.

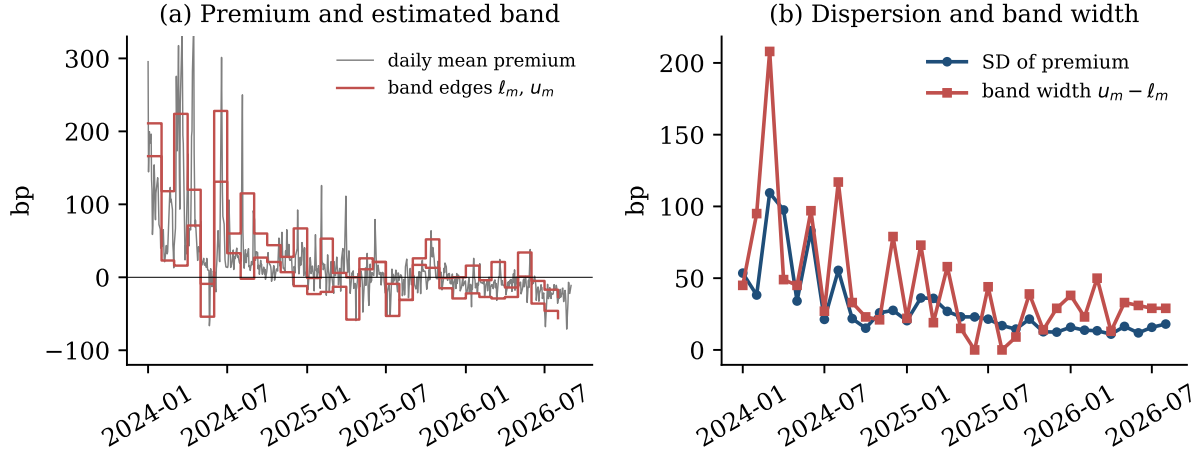


Figure 6: The USDT/TRY premium over the interbank rate. Panel (a): daily mean premium and the estimated monthly band edges ℓ_m and u_m . Panel (b): monthly standard deviation of the five-minute premium and the width of the estimated band.

6.2 Threshold dynamics

Table 2 reports the main estimates. Column (1) is a linear error-correction model with month fixed effects. It finds slow, uniform reversion ($\hat{\kappa} = 0.0039$ per five minutes). Column (2) estimates the threshold model. Outside the band the premium reverts at $\hat{\kappa}^+ = 0.0054$ above it and $\hat{\kappa}^- = 0.0085$ below it, both significant at 1%. Column (3) adds a linear reversion term that is active only *inside* the band. Its coefficient is -0.0006 with a standard error of 0.0007 , so there is no detectable mean reversion inside the band, as Proposition 2(a) predicts. The median band width is 32 bp, and 39% of intervals lie inside the band. A sup-LR test of linearity has limited power month by month: it rejects at 5% in 19% of months. Summed across months, however, the statistic is 464, and a day-clustered wild bootstrap under the linear null gives a p -value of 0.01, the smallest attainable with 99 draws.

Table 3 and Figure 7 look at reversion non-parametrically, by size of dislocation. Inside the band, proportional reversion is zero or slightly negative. Just outside the band (excess below 10 bp) there is no reversion either, and the premium drifts further out on average. This is what the model predicts when order flow is persistent: once flow pushes the premium through the edge of the band, it keeps moving toward the steady state $b^* = c + \beta q / (\lambda K)$ of Proposition 3. Order flow on other lira venues, which we do not observe, also contributes. Beyond 10 bp, reversion switches on: the proportional speed is 0.0052–0.0058 per five minutes, a half-life of about ten hours. For the very largest dislocations (more than 40 bp outside the band, mostly in the first half of 2024), reversion is slower again (0.0037). Proposition 2(c) alone does not predict this. It is, however, what Proposition 5 predicts when

Table 2: Threshold arbitrage in the USDT/TRY premium

	(1)	(2)	(3)	(4)
OFI_t	0.603*** (0.030)	0.604*** (0.030)	0.604*** (0.030)	0.635*** (0.032)
$-(b_{t-1} - u_m)_+ [\kappa^+]$		0.0054*** (0.0013)	0.0054*** (0.0013)	0.0025* (0.0013)
$(\ell_m - b_{t-1})_+ [\kappa^-]$		0.0085*** (0.0025)	0.0085*** (0.0025)	0.0074*** (0.0018)
$-b_{t-1} [\text{linear } \kappa]$	0.0039*** (0.0006)			
$-(b_{t-1} - m_m) \cdot \mathbb{I}\{\text{in band}\}$			-0.0006 (0.0007)	
$OFI_t \times BH$				-0.052 (0.033)
$-(b_{t-1} - u_m)_+ \times BH$				0.0079*** (0.0014)
$(\ell_m - b_{t-1})_+ \times BH$				0.0023 (0.0018)
Month fixed effects	Yes	Yes	Yes	Yes
Month-specific band $[\ell_m, u_m]$	No	Yes	Yes	Yes
N	162,624	162,624	162,624	162,624
Within R^2	0.042	0.042	0.042	0.043

Notes: Dependent variable: five-minute change in the premium (bp). OFI is standardized within month. $[\ell_m, u_m]$ is the band estimated month by month; m_m is its midpoint. BH is an indicator for Istanbul business hours (09:00–17:00). Standard errors clustered by day in parentheses. ***, **, * denote significance at 1%, 5%, 10%.

large and persistent dislocations coincide with depleted arbitrage capacity. We read it as suggestive evidence of capacity depletion rather than as a clean test.

6.3 Capacity states: banking hours

The model’s sharpest comparative static concerns capacity, and the trading day provides a direct source of variation. During Istanbul business hours, banks process transfers, treasury desks are staffed and the fiat leg of the arbitrage can be executed. Outside business hours, capacity falls even though the crypto market keeps trading. Column (4) of Table 2 interacts the reversion terms and order flow with an indicator for 09:00–17:00 Istanbul time. Above the band, the speed of arbitrage is 0.0025 outside business hours and 0.0104 during them. The difference, 0.0079 (s.e. 0.0014), is a fourfold increase. The primitive impact of order flow is similar across the two states (0.64 and 0.58). By Proposition 3, the long-run price impact of a sustained one-standard-deviation order-flow imbalance is therefore $\beta/\kappa^+ \approx 257$ bp outside business hours and 56 bp during them. The same demand moves the premium almost five times as much when arbitrage capacity is low. This is Corollary 1 within a single

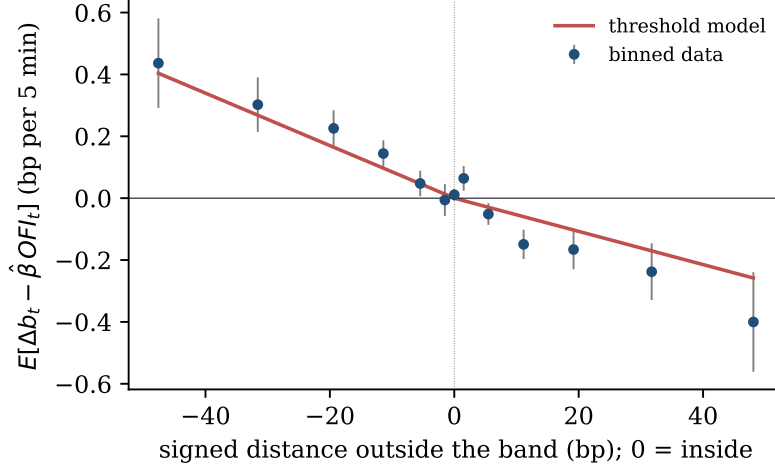


Figure 7: Expected premium change by position relative to the band. Mean five-minute change in the premium net of the order-flow component, by signed distance outside the month-specific band (0 = inside the band), net of month fixed effects, with 95% confidence intervals clustered by day. Dislocations more than 60 bp outside the band are omitted for readability. The red line is the fitted drift of column (2) of Table 2.

market. Below the band, the business-hours difference is positive but not significant.

Figure 8 shows the same pattern in levels. The mean absolute deviation of the premium from its daily median is lowest in the early afternoon in Istanbul, about 7 bp. It is roughly twice as large late in the evening, 13–15 bp, when stablecoin volume is still about two thirds of its daytime peak. The local market remains active after banks close; what disappears is the capacity to move money across the fiat leg.

6.4 Integration over time

Table 4 estimates (8) separately by year. The primitive price impact of standardized order flow falls from 1.04 bp in 2024 to 0.43 in 2025 and 0.25 in 2026. The speed of arbitrage above the band roughly doubles, from 0.0051 to 0.0095. The standard deviation of the premium falls from 74 to 16 bp. Through the lens of Proposition 4, the tails of the premium distribution, governed by $\lambda K/\sigma^2$, tightened as capacity grew. Over these two and a half years the lira stablecoin market became markedly more integrated with the interbank market.

6.5 Who discovers the dollar?

Proposition 6 says that the share ω of an arbitrage correction borne by the global price determines how much local information reaches it. We estimate ω by running (8) separately for the change in the log local price and the change in the log interbank rate (Table 5).

Table 3: Reversion by size of dislocation

	Mean distance (bp)	Reversion per 5 min, $\hat{\rho}$	Half-life (min)	N
Inside band, inner half	6.9	−0.0028 (0.0020)	∞	32,221
Inside band, outer half	21.6	−0.0020 (0.0006)	∞	31,020
Outside, excess $\in (0, 5]$ bp	2.5	−0.0154 (0.0055)	∞	20,099
Outside, excess $\in (5, 10]$ bp	7.4	0.0004 (0.0023)	7868	18,562
Outside, excess $\in (10, 20]$ bp	14.5	0.0052 (0.0011)	662	24,682
Outside, excess $\in (20, 40]$ bp	28.4	0.0058 (0.0008)	593	19,918
Outside, excess > 40 bp	81.9	0.0037 (0.0006)	949	16,122

Notes: Proportional reversion $\hat{\rho}$ is the sum of reversion toward the band (net of $\hat{\beta} OFI_t$) divided by the sum of distances. Distance is measured from the band midpoint for observations inside the band and from the nearest edge for observations outside it. Delta-method standard errors clustered by day in parentheses. Half-lives are in minutes.

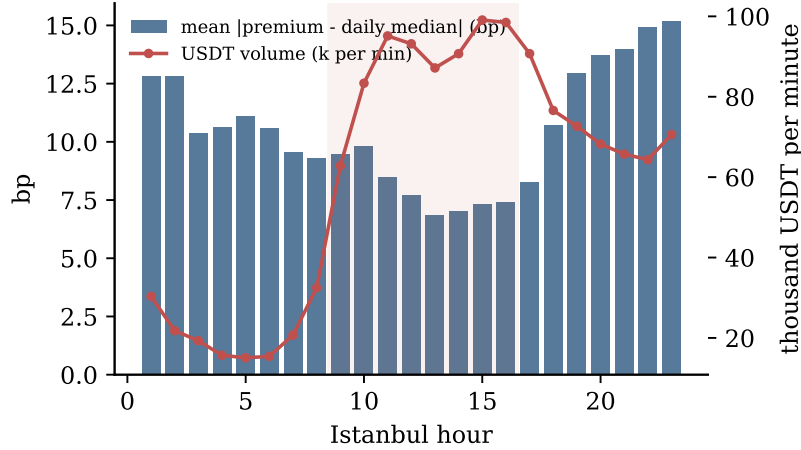


Figure 8: Intraday pattern. Bars: mean absolute deviation of the one-minute premium from its daily median, by Istanbul hour (weekdays). Line: mean USDT/TRY volume per minute. The shaded area marks business hours.

Essentially all of the correction happens in the stablecoin market. When the premium is above the band, the local price falls at rate 0.0051 while the interbank rate does not move (−0.0002, s.e. 0.0004), so $\hat{\omega} = 0.04$. Below the band the point estimate is 0.28, but the global-leg coefficient is not significant. Local order flow moves the local price one-for-one with the premium (0.61) and has no effect on the interbank rate (0.003, s.e. 0.018). In the language of the title, the dollar’s lira price is discovered in the interbank market. The stablecoin market follows it, with a lag governed by arbitrage capacity, and local information that arrives first in the stablecoin market is not impounded into the interbank rate.

The 19 March 2025 shock. The detention of Istanbul’s mayor on the morning of 19 March 2025 triggered one of the largest intraday moves in the lira’s history. Figure 9 shows

Table 4: Estimates by year

Year	β	κ^+	κ^-	Median band width (bp)	SD of premium (bp)
2024	1.041*** (0.055)	0.0051*** (0.0014)	0.0050*** (0.0011)	47	74.1
2025	0.431*** (0.040)	0.0065*** (0.0015)	0.0212** (0.0091)	20	24.5
2026	0.249*** (0.018)	0.0095*** (0.0015)	0.0094*** (0.0020)	30	16.4

Notes: Specification (8) with year-specific slopes, month fixed effects and month-specific bands. Standard errors clustered by day.

Table 5: Which price adjusts?

	$\Delta \log P_t^L$ (USDT/TRY)	$\Delta \log P_t^G$ (USD/TRY)
OFI _t	0.609*** (0.030)	0.003 (0.018)
$-(b_{t-1} - u_m)_+$	0.0051*** (0.0012)	-0.0002 (0.0004)
$(\ell_m - b_{t-1})_+$	0.0062*** (0.0009)	-0.0024 (0.0023)
Month fixed effects	Yes	Yes
Implied global share ω (premium side)		0.04
Implied global share ω (discount side)		0.28
N	162,624	162,624

Notes: Specification (8) with the five-minute change in the log USDT/TRY price (column 1) or the log interbank USD/TRY rate (column 2), in bp, as the dependent variable. ω is the global price's share of the total correction, $\kappa_G/(\kappa_L + \kappa_G)$. Standard errors clustered by day.

the minute-by-minute response. The interbank rate jumped by up to 11.0%. USDT/TRY rose by at most 6.0%, so the premium fell to -660 bp at 07:46 UTC, far outside the March band. Consistent with the evidence above, the stablecoin market did not lead the move. The dislocation then closed from both sides: the interbank rate retraced amid heavy official intervention, and the stablecoin price rose. The premium was back inside the band within 75 minutes. By the afternoon both prices were about 3–3.5% above their pre-shock levels. The episode illustrates Proposition 2(b). A dislocation far outside the band is corrected quickly, and the correction is fastest when the dislocation is largest.

Order-flow identification. Section 5 argued that in-band observations identify the primitive impact β free of arbitrage trades. In the data, $\hat{\beta}$ estimated on in-band observations (0.56, s.e. 0.03) is close to, and slightly below, the out-of-band estimate (0.64). Arbitrage trades therefore do not appear to contaminate Binance taker flow materially. This is plausible if arbitrageurs execute the local leg on other venues or over the counter.

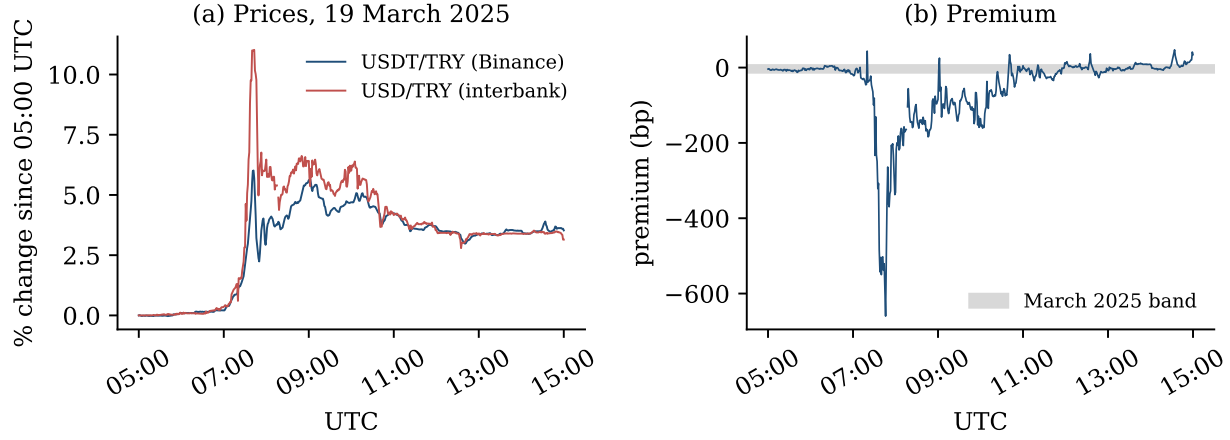


Figure 9: The 19 March 2025 lira shock. Panel (a): percentage change in USDT/TRY (Binance VWAP) and interbank USD/TRY since 05:00 UTC. Panel (b): one-minute premium and the band estimated for March 2025.

6.6 Robustness

Table 6 re-estimates the main specification at one- and fifteen-minute frequencies, with last-trade prices instead of VWAPs, excluding the 19 March – 30 April 2025 window, with quarterly instead of monthly bands, and with 10% trimming. The reversion speeds outside the band are positive and significant in every specification. They scale roughly in proportion to the sampling interval, as they should: the fifteen-minute estimates are about three times the five-minute ones. The business-hours increase in κ^+ is significant at 1% throughout. Inside-band reversion is statistically zero in four of the six alternatives. It is small but positive with quarterly bands and at the fifteen-minute frequency, where it is one fifth or less of the outside-band speed. Coarser bands mechanically put some out-of-band observations inside the band, which is the expected direction of that bias.

Limitations. Three caveats apply. First, we observe order flow on one venue; flow on other lira venues and over the counter is part of the residual. Second, the interbank benchmark is an offshore quote; at night it is thinner, which could add noise to the premium outside business hours. However, its median spread is about 1 bp, so the fourfold difference in arbitrage speed is unlikely to be a quoting artifact. Third, the band is identified from the dynamics of the premium alone. Direct measures of arbitrage capacity, such as exchange withdrawal data or bank transfer limits, would allow a sharper test of Propositions 5 and 7.

Table 6: Robustness

	β	κ^+	κ^-	In-band reversion	$\kappa_{BH}^+ - \kappa_{off}^+$
Baseline: 5-min, VWAP, monthly bands, 5% trimming	0.604*** (0.030)	0.0054*** (0.0013)	0.0085*** (0.0025)	-0.0006 (0.0007)	0.0079*** (0.0014)
1-minute frequency	0.342*** (0.013)	0.0011*** (0.0003)	0.0036*** (0.0009)	-0.0005* (0.0003)	0.0014*** (0.0003)
15-minute frequency	1.116*** (0.067)	0.0187*** (0.0028)	0.0243*** (0.0070)	0.0039** (0.0018)	0.0278*** (0.0045)
Last-trade price instead of VWAP	0.643*** (0.032)	0.0056*** (0.0013)	0.0091*** (0.0025)	-0.0006 (0.0007)	0.0080*** (0.0015)
Excluding 19 Mar–30 Apr 2025	0.598*** (0.029)	0.0054*** (0.0013)	0.0062*** (0.0010)	-0.0005 (0.0007)	0.0080*** (0.0015)
Quarterly bands	0.604*** (0.030)	0.0044*** (0.0011)	0.0130*** (0.0046)	0.0017*** (0.0006)	0.0062*** (0.0015)
10% trimming	0.604*** (0.030)	0.0042*** (0.0011)	0.0098*** (0.0034)	-0.0002 (0.0007)	0.0070*** (0.0014)

Notes: Each row re-estimates columns (2)–(4) of Table 2. “In-band reversion” is the coefficient on $-(b_{t-1} - m_m)\mathbb{I}\{\text{in band}\}$ in column (3). The last column is the business-hours interaction on κ^+ in column (4). Reversion coefficients are per sampling interval. Standard errors clustered by day.

7 Simulation evidence

This section shows that the estimator recovers the structural parameters at realistic sample sizes and that linear models are misleading when the data are generated by threshold arbitrage. Units are basis points and minutes. The baseline design has $c^\pm = 30$ bp, $\kappa^\pm = 0.05$ per minute (half-life 14 minutes), $\beta = 1$, $\sigma = 4$ bp/ $\sqrt{\text{min}}$, and order flow following a standardized AR(1) with autocorrelation 0.9. The asymmetric design has $c^+ = 20$, $c^- = 50$, $\kappa^+ = 0.08$ and $\kappa^- = 0.02$. The low-capacity design sets $\kappa^\pm = 0.015$. Each replication has $T = 20,000$ observations, about two weeks of minute data, and there are 200 replications per design. Thresholds are profiled over a grid with 2.5 bp spacing.

Table 7 reports the results. The thresholds are estimated with an RMSE of 2–3 bp in the baseline. Capacity-scaled speeds and order-flow impact are essentially unbiased. Asymmetries in both the band and capacity are recovered. In the low-capacity design the premium spends more time far outside the band, so its edges are visited less sharply and the thresholds are estimated less precisely, though still without bias. The implied long-run impact multipliers β/κ^+ have median estimates of 19.8, 12.3 and 63.9 against true values of 20.0, 12.5 and 66.7. The capacity differences that Corollary 1 says are hidden in raw premia are therefore recoverable from the joint dynamics of premia and order flow.

The linear error-correction model performs poorly. Its reversion coefficient is 0.015 in the baseline, less than a third of the true speed outside the band, and it implies a half-life of 47 minutes instead of 14. In the asymmetric design it averages two very different speeds into a single number. Reading the linear coefficient as a measure of arbitrage capacity, and $\beta/\hat{\kappa}$ as long-run impact, would overstate the long-run price impact of order flow by a factor of

Table 7: Monte Carlo: recovery of structural parameters

	c^+	c^-	$\kappa^+ = \lambda K^+$	$\kappa^- = \lambda K^-$	β
<i>Symmetric design</i>					
True	30.0	30.0	0.050	0.050	1.000
Median estimate	30.0	30.0	0.050	0.051	0.999
RMSE	2.5	2.7	0.007	0.007	0.029
Linear ECM $\hat{\kappa}$			0.0147		0.964
<i>Asymmetric design</i>					
True	20.0	50.0	0.080	0.020	1.000
Median estimate	20.0	50.0	0.081	0.020	0.999
RMSE	2.0	6.8	0.011	0.004	0.028
Linear ECM $\hat{\kappa}$			0.0105		0.963
<i>Low capacity design</i>					
True	30.0	30.0	0.015	0.015	1.000
Median estimate	30.0	30.0	0.016	0.015	0.999
RMSE	8.5	8.9	0.004	0.004	0.028
Linear ECM $\hat{\kappa}$			0.0075		0.991

Notes: 200 replications of $T = 20,000$ one-minute observations per design. “Median estimate” and “RMSE” refer to the profiled least-squares estimator of (7). “Linear ECM” reports the median reversion coefficient (first four columns) and order-flow coefficient (last column) from $\Delta b_{t+1} = \alpha + \beta OFI_t - \kappa b_t + \varepsilon_{t+1}$.

about two (low-capacity design) to more than three (baseline).

Table 8 implements test T2 on a long simulated path. Reversion is absent inside the band: the estimated slopes are economically negligible (at most 0.003 in absolute value, and not positive), and the implied half-lives are infinite. It switches on at the band edge and strengthens with the size of the dislocation, with the half-life falling from 68 minutes for dislocations just outside the band to 25 minutes for the largest ones. This pattern is the empirical signature of Proposition 2(c), and it is inconsistent with any linear model.

8 Conclusion

The law of one price for dollar stablecoins holds only as far as arbitrage capacity allows. Two primitive frictions, a proportional cost and a finite capacity, turn the local stablecoin premium into a threshold diffusion with a closed-form stationary law. The same frictions separate what trading volume reveals (demand) from what the premium reveals (capacity). They make long-run price impact convex and bounded by an arbitrage throughput limit. They filter which local information reaches the global price, and they make centralized arbitrage networks both efficient and fragile.

Table 8: Reversion speed by size of dislocation (baseline design)

$ b_t $ (bp)	[0,10)	[10,20)	[20,30)	[30,45)	[45,60)	[60,90)
Reversion speed per minute	-0.0027	-0.0011	-0.0003	0.0101	0.0212	0.0273
Implied half-life (min)	∞	∞	∞	68	33	25
Share of observations	0.175	0.182	0.190	0.271	0.143	0.039

Notes: Simulated path of 5×10^5 minutes. Within each bin of $|b_t|$, the table reports the slope from regressing $-\Delta b_{t+1} \text{ sign}(b_t)$ on $|b_t|$ and $OFI_t \text{ sign}(b_t)$ without an intercept, and the implied half-life $\ln 2/\text{slope}$ (reported as ∞ when the slope is not positive). The band is $[-30, 30]$ bp.

The lira stablecoin market bears these predictions out. Premia inside a band of a few tens of basis points do not revert. Arbitrage switches on outside the band and is four times faster when banks are open. The stablecoin market does the adjusting while the interbank market sets the price, and the market has integrated rapidly as capacity has grown. Natural next steps are to measure capacity directly, using exchange withdrawal flows and bank transfer limits, and to compare markets with different capital-control regimes, such as the naira, peso and won markets, where the model predicts different bands for the same demand. Both would sharpen the answer to the question in the title: whether a local market helps discover the dollar’s value, or only pays to access it.

Data and code availability. All data used in this paper are public: Binance spot candles (<https://data.binance.vision>) and Dukascopy interbank candles. Code that downloads the data and reproduces every table and figure is available from the authors.

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Appendix: Proofs

Proof of Proposition 1. $\Pi(\cdot; b)$ is continuous and strictly concave on $\mathbb{R}$: it is strictly concave on each half-line, and at $a = 0$ its left derivative $b + c^-$ exceeds its right derivative $b - c^+$, so the kink is concave. Hence the maximizer is unique and is characterized by $0 \in \partial\Pi(a^*; b)$. At $a = 0$ the superdifferential is $[b - c^+, b + c^-]$, which contains 0 if and only if $-c^- \leq b \leq c^+$. If $b > c^+$, the right derivative at zero is positive, so $a^* > 0$ solves $b - c^+ - a/K^+ = 0$, i.e. $a^* = K^+(b - c^+)$. If $b < -c^-$, the left derivative at zero is negative, so $a^* < 0$ solves $b + c^- - a/K^- = 0$, i.e. $a^* = K^-(b + c^-) = -K^-(-b - c^-)$. Combining the cases gives (2). $\square$

Proof of Proposition 2. (a) With $q \equiv 0$ and $b_s \in (-c^-, c^+)$ for $s < \tau$, the drift vanishes, so $b_{t \wedge \tau} = b_0 + \sigma W_{t \wedge \tau}$ is a stopped Brownian motion and hence a martingale. The expected exit time of $b_0 + \sigma W$ from $(-c^-, c^+)$ solves $\frac{\sigma^2}{2} u'' = -1$ with $u(-c^-) = u(c^+) = 0$, whose solution is $u(b_0) = (c^+ - b_0)(b_0 + c^-)/\sigma^2$. (b) With $\sigma = 0$ and $b > c^+$, $\dot{b} = -\kappa^+(b - c^+)$. The solution $b_t = c^+ + (b_0 - c^+)e^{-\kappa^+ t}$ stays above c^+ for all t , so the linear dynamics remain valid. The excess halves when $e^{-\kappa^+ t} = 1/2$. (c) For $b > c^+$, $\rho(b) = \kappa^+(b - c^+)/b = \kappa^+(1 - c^+/b)$, which is increasing in b and tends to κ^+ . On the band $\rho = 0$. The case $b < -c^-$ is symmetric. $\square$

Proof of Proposition 3. (a) Let $q > 0$ and $\sigma = 0$, so $\dot{b} = f(b) \equiv \beta q + \mu(b)$. The function f is continuous and strictly positive for $b \leq c^+$, strictly decreasing on (c^+, ∞) , and has a unique zero at $b^* = c^+ + \beta q/\kappa^+$. Since $f > 0$ below b^* and $f < 0$ above it, every trajectory converges monotonically to b^* . The case $q < 0$ is symmetric. If $q = 0$, f vanishes on the band and the argument of Proposition 2(b) applies outside it. (b) Differentiate b^* . (c) At the steady state $\dot{b} = 0$, so $\beta q = \lambda a^*$. $\square$

Proof of Corollary 1. Immediate from Proposition 3(a) applied to each market. $\square$

Proof of Proposition 4. For a one-dimensional diffusion $db = \mu(b)dt + \sigma dW$ with Lipschitz drift, the speed measure has density $m(b) = \frac{2}{\sigma^2} \exp\left(\frac{2}{\sigma^2} \int_0^b \mu(x)dx\right)$. The process is positive recurrent, with stationary density proportional to m , whenever m is integrable and the scale function diverges at $\pm\infty$. Here $\int_0^b \mu = 0$ on the band, $-\frac{\kappa^+}{2}(b - c^+)^2$ for $b > c^+$, and $-\frac{\kappa^-}{2}(b + c^-)^2$ for $b < -c^-$. This gives (4), which is integrable. The scale density $1/m$ grows like a Gaussian reciprocal, so the scale function diverges at both ends. For the constant, use $\int_0^\infty e^{-\theta x^2} dx = \frac{1}{2} \sqrt{\pi/\theta}$ with $\theta = \kappa^\pm/\sigma^2$. Part (a) follows by integrating the plateau. For part (b), $\int_{-c^-}^{c^+} b db = \frac{(c^+)^2 - (c^-)^2}{2}$ and $\int_{c^+}^\infty b e^{-\theta(b - c^+)^2} db = c^+ \cdot \frac{1}{2} \sqrt{\pi/\theta} + \frac{1}{2\theta}$, with $\frac{1}{2\theta} = \frac{\sigma^2}{2\kappa^+}$; the lower tail is symmetric with the opposite sign. $\square$

Proof of Proposition 5. Write $x_t = b_t - c$ and $a_t = K_t(x_t)_+ \geq 0$. The system is

$$\dot{x} = \beta q - \lambda a, \quad \dot{K} = \xi(\bar{K} - K) - \chi a.$$

The set $\{K > 0\}$ is forward invariant: at $K = 0$, $a = 0$ and $\dot{K} = \xi\bar{K} > 0$.

(a) A steady state requires $\lambda a = \beta q > 0$, so $x > 0$ and $a = Kx = \beta q/\lambda$. Then $\dot{K} = 0$ gives $K^* = \bar{K} - \chi\beta q/(\xi\lambda) = \bar{K}(1 - q/q_{crit})$, which is positive if and only if $q < q_{crit}$, and $x^* = \beta q/(\lambda K^*)$. In the region $x > 0$ the Jacobian is

$$J = \begin{pmatrix} -\lambda K & -\lambda x \\ -\chi K & -\xi - \chi x \end{pmatrix}, \quad \text{tr } J < 0, \quad \det J = \lambda K(\xi + \chi x) - \lambda\chi Kx = \lambda\xi K > 0,$$

so both eigenvalues have negative real parts at (x^*, K^*) , which establishes local asymptotic stability. Differentiating $b^* = c + \beta q/(\lambda\bar{K}(1 - q/q_{crit}))$ gives $\partial b^*/\partial q = \beta/(\lambda\bar{K}) \cdot [(1 - q/q_{crit}) + q/q_{crit}]/(1 - q/q_{crit})^2 = \beta/(\lambda\bar{K}(1 - q/q_{crit})^2)$, which is increasing and convex on $[0, q_{crit})$ and diverges as $q \uparrow q_{crit}$.

(b) Integrating the capacity equation over $[0, T]$ and using $K_T > 0$ and $K_t \geq 0$,

$$\chi \int_0^T a_t dt = \xi\bar{K}T - \xi \int_0^T K_t dt - (K_T - K_0) \leq \xi\bar{K}T + K_0.$$

Hence $\int_0^T a_t dt \leq \bar{a}T + K_0/\chi$. Integrating the premium equation,

$$x_T = x_0 + \beta qT - \lambda \int_0^T a_t dt \geq x_0 - \frac{\lambda K_0}{\chi} + (\beta q - \lambda\bar{a})T.$$

Dividing by T and letting $T \rightarrow \infty$ gives $\liminf x_T/T \geq \beta q - \lambda\bar{a} = \beta(q - q_{crit})$. If $q > q_{crit}$ this is positive, so no steady state can exist. $\square$

Proof of Proposition 6. After the shock, $b_0 = \delta$ and $\dot{b} = -\lambda a^*(b)$, with $\dot{p}^G = \lambda_G a^*(b)$. If $\delta \leq c^+$, $a^* = 0$ for all t and p^G does not move. If $\delta > c^+$, Proposition 2(b) gives $b_t - c^+ = (\delta - c^+)e^{-\lambda K^+ t}$. Since $\dot{p}^G = \lambda_G a^* = -\frac{\lambda_G}{\lambda} \dot{b}$,

$$p_h^G - p_0^G = \omega(b_0 - b_h) = \omega(\delta - c^+)(1 - e^{-\lambda K^+ h}).$$

Dividing by δ gives $S_h(\delta)$, and parts (a)–(c) follow by inspection. $\square$

Proof of Proposition 7. Since P is a projection and $P\mathbf{1} = 0$, $z = Pp$ satisfies $dz = -PF(p)dt + P\Sigma dW$, where $F_i(p) = \sum_j w_{ij}\phi_{c_{ij}}(p_i - p_j)$. By Itô's lemma, $V(z) = \frac{1}{2}\|z\|^2$ has

generator

$$\mathcal{L}V = -z'F(p) + \frac{1}{2} \text{tr}(P\Sigma\Sigma'P),$$

where we used $z'P = z'$. By symmetry of w and oddness of ϕ ,

$$z'F(p) = \sum_i \sum_j w_{ij} z_i \phi_{c_{ij}}(p_i - p_j) = \sum_{i < j} w_{ij} (z_i - z_j) \phi_{c_{ij}}(p_i - p_j) = \sum_{i < j} w_{ij} (p_i - p_j) \phi_{c_{ij}}(p_i - p_j),$$

since $z_i - z_j = p_i - p_j$. For any x and $c > 0$, $x\phi_c(x) \geq \frac{1}{2}x^2 - \frac{1}{2}c^2$. This holds trivially for $|x| \leq c$; for $|x| > c$, $x\phi_c(x) = x^2 - c|x| \geq x^2 - \frac{1}{2}(x^2 + c^2)$. Therefore

$$z'F(p) \geq \frac{1}{2} \sum_{i < j} w_{ij} (p_i - p_j)^2 - \frac{1}{2} \sum_{i < j} w_{ij} c_{ij}^2 = \frac{1}{2} z' L_w z - \frac{1}{2} \sum_{i < j} w_{ij} c_{ij}^2.$$

Since $z \perp \mathbf{1}$, the Courant–Fischer theorem gives $z' L_w z \geq \lambda_2(L_w) \|z\|^2 = 2\lambda_2(L_w)V$. Thus

$$\mathcal{L}V \leq -\lambda_2(L_w)V + C, \quad C \equiv \frac{1}{2} \sum_{i < j} w_{ij} c_{ij}^2 + \frac{1}{2} \text{tr}(P\Sigma\Sigma'P).$$

By Dynkin's formula with a standard localization argument (the drift is Lipschitz, so all moments are finite), $\frac{d}{dt} \mathbb{E}V(z_t) \leq -\lambda_2 \mathbb{E}V(z_t) + C$. Grönwall's inequality then gives $\mathbb{E}V(z_t) \leq e^{-\lambda_2 t} \mathbb{E}V(z_0) + C/\lambda_2$, which yields the bound. If the network is disconnected, take two components A and B . The difference between their average prices has no drift, because arbitrage within each component preserves its average. If the noise is non-degenerate across components, that difference is a Brownian motion with positive variance rate, so its second moment, and hence $\mathbb{E}V$, grows linearly. $\square$

Proof of Corollary 2. (a) The Laplacian of the star $K_{1,N-1}$ with weight w has eigenvalues 0, w (multiplicity $N-2$) and Nw , so $\lambda_2 = w$. For any non-complete graph, Fiedler (1973) shows that algebraic connectivity is bounded above by vertex connectivity. Every tree on $N \geq 3$ vertices has vertex connectivity one, so $\lambda_2 \leq w$ for every spanning tree with weight w , and the star attains the bound. (b) Deleting the hub removes all edges. Deleting a leaf leaves a star on $N-1$ vertices, whose algebraic connectivity is again w for $N-1 \geq 3$ (for $N-1=2$ it is $2w$). Unbounded dispersion follows from the last part of Proposition 7. (c) The complete graph K_N with weight w has Laplacian $w(NI - \mathbf{1}\mathbf{1}')$, with eigenvalues 0 and Nw (multiplicity $N-1$). Removing a vertex leaves K_{N-1} . $\square$